

Submission LINALG1-QUIZ-SUB01

Question LINALG1-QUIZ-Q01

scratch: show step

1. $\det = 6(2) - 4(4) = -4$
 2. Eliminate y , yielding $x=1$; substitute $\Rightarrow y=2$
 3. checks: $6(1)+4(2)=14$, $4(1)+2(2)=8$
- $\Rightarrow$ FINAL: $x=1$, $y=2$ (check units/interpretation)

Question LINALG1-QUIZ-Q02

scratch: show step

1. $\det(A) = 2 \cdot 5 - 2 \cdot 2 = 6$
 2. $\det(A) \neq 0$, so A is invertible.
- $\Rightarrow$ FINAL: $\det=6$; invertible (check units/interpretation)

Question LINALG1-QUIZ-Q03

scratch: show step

1. $T(1, -1) = (1 - 2, 3 + 1) = (-1, 4)$
 2. Matrix $\begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$
 3. Both coordinate rules have no constant term, so T is linear.
- $\Rightarrow$ FINAL: $T(1, -1) = (-1, 4)$; matrix $\begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$; linear (check units/interpretation)

Question LINALG1-QUIZ-Q04

scratch: show step

1. Set $c_1(1, 0) + c_2(0, 1) = (0, 0)$.
 2. Coordinates give $c_1 = 0$ and $c_2 = 0$ only, so the set is independent.
- $\Rightarrow$ FINAL: only trivial combination; $\{(1, 0), (0, 1)\}$ is linearly independent (check units/interpretation)

Question LINALG1-QUIZ-Q05

scratch: show step

1. $\text{Null}(A)$ is the set of vectors sent to the zero vector.
 2. It is the solution space of $Ax=0$ and is a subspace.
- $\Rightarrow$ FINAL: the vectors mapped to zero; exactly the solution space of $Ax=0$ (check units/interpretation)